Surveillance of High Pathogenicity Avian Influenza for Smallholder Poultry Systems in Resource-Limited Settings

In accordance with the WOAH Terrestrial Animal Health Standards

World Organisation for Animal Health

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List of abbreviations

AI	Avian Influenza
AIV	Avian Influenza Virus
CAHWs	Community Animal Health Workers
ELISA	Enzyme-Linked Immunosorbent Assay
FAO	Food and Agriculture Organization of the United Nations
HPAI	High Pathogenicity Avian Influenza
HPAIV	High Pathogenicity Avian Influenza Virus
H5Nx Gs/GD	Lineage H5Nx Goose/Guangdong of the HPAIV
PDS	Participatory Disease Surveillance
PCR	Polymerase Chain Reaction
PPE	Protective Personal Equipment
PPPs	Public–Private Partnerships
SHPS	Smallholder Poultry Systems
WOAH	World Organisation for Animal Health
VA	Veterinary Authority
VPPs	Veterinary Paraprofessionals
VS	Veterinary Services

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Glossary

Case
An individual animal infected by a pathogenic agent, with or without showing clinical signs.

Disease
The clinical and pathological manifestation of infection.

Epidemiological unit
A group of animals with the same likelihood of exposure to a pathogenic agent. In some circumstances, the epidemiological unit may consist of a single animal.

Infection
The entry and development or multiplication of an infectious agent in the body of humans or animals.

Outbreak
The occurrence of one or more cases within an epidemiological unit.

Population
A group of units sharing a common defined characteristic in a particular geographical area at a given time.

Poultry
For the purposes of these guidelines, *poultry* specifically refers to birds kept in commercial farms and those maintained within a single household for personal use, without direct or indirect contact with commercial poultry or their facilities. It also includes birds kept for various purposes, including shows, racing, exhibitions, competitions, or breeding and selling for these purposes, and pet birds. (Adapted from the WOAH *Terrestrial Animal Health Code* Glossary).

Poultry production system
An organised and interrelated set of components designed to produce poultry and poultry-related products.

Prevalence
The total number of cases or outbreaks of a disease present in a population at risk, in a particular geographical area, at a specified time or over a defined period.

Risk
The likelihood of occurrence and the potential magnitude of the biological and economic consequences of an adverse event or effect to animal or human health.

Sampling unit
The unit selected for sampling in a study or during surveillance activities. This may be an individual animal or a group of animals (e.g. an epidemiological unit).

Sensitivity
The proportion of truly positive units that are correctly identified as positive by a test.

Specificity
The proportion of truly negative units that are correctly identified as negative by a test.

Study (sampling) population
The population from which surveillance data are derived. This may be the same as, or a subset of, the target population.

Surveillance
The systematic and continuous collection, collation and analysis of information related to animal health, and the timely dissemination of this information to those who need to know so that action can be taken.

Surveillance activity
A single activity or action undertaken with the primary purpose of collecting, collating, analysing or disseminating information.

Surveillance programme
A set of surveillance activities used to investigate the occurrence of one or more hazards in a specified population. A surveillance programme usually forms part of a broader surveillance system.

Surveillance system
A collection of surveillance components (programmes) that generate information on the health and disease status of animal populations.

Survey
An investigation in which information is collected systematically, usually carried out on a sample of a defined population group and within a defined time period.

Target population
The population about which conclusions are to be inferred.

Value chain
The full range of activities required to bring a product or service from conception through the different production phases (involving a combination of physical transformation and the input of various producer services) and delivery to end consumers, including final disposal after use.

Veterinary Authority (VA)
The Governmental Authority of a WOAH Member comprising veterinarians, other professionals and paraprofessionals, with the responsibility and competence for ensuring or supervising the implementation of animal health and welfare measures, international veterinary certification and other standards and recommendations given in the *Terrestrial Animal Health Code* throughout the whole territory.

Veterinary Services (VS)
The combination of governmental and non-governmental individuals and organisations that carry out activities to implement WOAH standards in a Member state. The VS operate under the overall authority and direction of the VA. Private-sector organisations, veterinarians, veterinary paraprofessionals (VPPs), community animal health workers (CAHWs) and other relevant personnel may be accredited or approved by the VA to perform delegated functions.

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Summary

This publication represents a collaborative effort led by the World Organisation for Animal Health (WOAH), in conjunction with animal health experts from the Jockey Club College of Veterinary Medicine and Life Sciences Research Centre for Applied One Health Research and Policy Advice of the City University of Hong Kong, the Agriculture, Fisheries and Conservation Department of Hong Kong Special Administrative Region of the People's Republic of China and the Institut National de Recherche pour l'Agriculture, l'Alimentation et l'Environnement, as well as experts from around the globe, to compile scientific evidence, field experience and professional insight for improving the health of smallholder poultry systems (SHPS) in resource-limited settings. Enhancing SHPS health not only benefits farmers' livelihoods but also supports family welfare and boosts community resilience.

Low- and middle-income countries (LMIC) of the world face several challenges in preventing and responding to avian influenza outbreaks, particularly among smallholder (backyard) poultry farmers. Low-sensitivity surveillance systems in SHPS hinder early detection, thereby increasing the spread of the virus. Furthermore, conventional surveillance programmes often demand substantial and ongoing funding while lacking the flexibility to address the full range of disease surveillance needs in SHPS, particularly where resources are constrained.

The Animal Health Forum on avian influenza hosted at the 90th WOAH General Session in May 2023 discussed these issues, leading to Resolution No. 28, which emphasises supporting smallholders with effective disease prevention measures. To this end, practical surveillance guidelines, tailored to local contexts, are essential for safeguarding farmers' livelihoods and ensuring the stability of the poultry industry. During the development of these guidelines, discussions among experts, WOAH Reference Centre experts and authors highlighted the feasibility and challenges of establishing a standardised methodology for such inherently complex and diverse production systems.

Given the constantly evolving nature of the high pathogenicity avian influenza virus, and the changing socio-cultural and economic context of SHPS around the world, this publication is intended as a living resource that will achieve its full value through contextual application, critical feedback and continuous evidence-informed revision. WOAH therefore encourages users of these guidelines to share their implementation experience and to provide constructive feedback through WOAH Delegates or their representatives, so that this publication can be further refined and adapted to the ever-changing ecology of highly pathogenic avian influenza viruses, food production systems and surveillance tools.

1. Background

Since the identification of the H5Nx Gs/GD lineage in 1996 [1], high pathogenicity avian influenza virus (HPAIV) has led to the culling of over 500 million poultry birds, affecting approximately 88 countries [2]. As zoonotic pathogens, these viruses have been associated with over 2,500 human cases linked to multiple strains, including H5N1, H7N9 and H5N6 since 2003 [3]. Since 2020, high pathogenicity avian influenza (HPAI) has spread rapidly to new territories, including South America. The resulting increase in poultry outbreaks, economic losses and unprecedented detections in mammals has confirmed HPAI as a disease of significant global concern [4].

In animal health, the primary objective of disease surveillance is to demonstrate the absence of infection or infestation, or to detect the presence or distribution of infection or infestation caused by exotic or emerging disease-causing agents as early as possible [5].

Disease surveillance is crucial to provide a better understanding on disease epidemiology, supporting the design of prevention and control measures adapted to local contexts.

The *Terrestrial Animal Health Code* by the World Organisation for Animal Health (WOAH) provides recommendations for designing, implementing and evaluating surveillance systems. It emphasises that such systems should be tailored to the specific context of each country and may comprise several surveillance programmes as components. These programmes must align with the main objectives of the national surveillance system while addressing and adapting to the specific needs and socio-economic realities of farms and communities.

1.1. Smallholder poultry systems in resource-limited settings

Smallholder poultry systems (SHPS), also referred to as backyard or village poultry systems, have unique features compared with commercial poultry production [6]. Typically located in rural and peri-urban areas, SHPS are characterised by basic infrastructure, flexible management practices [6,7] and the rearing of indigenous or mixed breeds that are well adapted to scavenging and often coexist with other animals [6,8]. By providing food and income, SHPS play a vital role in strengthening household economies and food security, and they serve as a safety net during times of hardship [9,10]. These systems also contribute manure for crop production, assist in pest control and hold cultural importance, for example through their role in traditional ceremonies [8,11]. For many rural communities, SHPS are essential for reducing poverty and empowering women and other vulnerable groups, ultimately improving livelihoods and fostering economic independence [12-14]. In these contexts, the concept of 'value' extends beyond economic (utilitarian) factors to include cultural, social, ecological and spiritual dimensions, which vary by region and community.

The socio-cultural and economic diversity of SHPS leads to wide variation in how disease prevention and control are understood and managed across countries. Farmers' approaches to managing disease risks differ considerably, particularly when confronted with challenges like emerging pathogens [15] and financial uncertainties [16-18]. These farming systems have distinct characteristics and serve other purposes than commercial poultry production. Typically, the role of SHPS in the poultry value chain is limited to local markets with modest impact on the national economy. Resources dedicated to the health and development of SHPS are also often scarce, particularly in low- and middle-income countries. This resource limitation creates several obstacles to implementing effective and sustainable disease surveillance programmes (**Box 1**).

BOX 1.

COMMON CHALLENGES OF DISEASE SURVEILLANCE IN SHPS IN RESOURCE-LIMITED SETTINGS

Limited data/information: there is insufficient knowledge regarding the number and locations of SHPS.

Delayed/underreporting from farmers: there is limited awareness about the specifics of disease aetiology and the zoonotic potential of HPAI, making it hard to differentiate from other infectious diseases of concern. Additionally, there is minimal support for farmers to report suspected cases, and they often do not know whom to contact.

Limited access to animal health services: there is a shortage of animal health service providers at the local level to collect samples and connect farmers with Veterinary Services.

Inadequate laboratory resources: there is a shortage of necessary resources for disease detection and testing, including trained personnel, infrastructure and reagents.

Inappropriate response to reporting: there is no feedback mechanism to farmers after they report cases. Culling is often the standard response to outbreaks and appropriate compensation schemes for affected farmers are often lacking.

1.2. A participatory approach to designing a sustainable surveillance programme

To ensure that a surveillance programme is cost-effective, practical and supported by local communities, it is critical to first establish clear communication and mutual trust among all key stakeholders [6]. One proven approach is participatory disease surveillance (PDS), a form of community-based surveillance that draws on local knowledge to determine whether a specific disease is present in a given area [19,20].

Using participatory rural appraisal methods such as informal semi-structured interviews, scoring, ranking and visual tools (e.g. participatory maps and timelines), PDS has proven valuable in detecting animal diseases, encouraging communication and collaboration, and building trust within local communities across diverse contexts (**Figure 1**). A key element of this approach is the early involvement of relevant stakeholders from diverse socio-cultural backgrounds, disciplines and economic sectors. Their input in describing community contexts, farming purposes and interactions within and beyond the value chain will help understand the complexities of local poultry production systems. This understanding is crucial for assessing the system's potential role in the ecology of HPAIV, and for designing sustainable and socially accepted surveillance programmes [6,21].

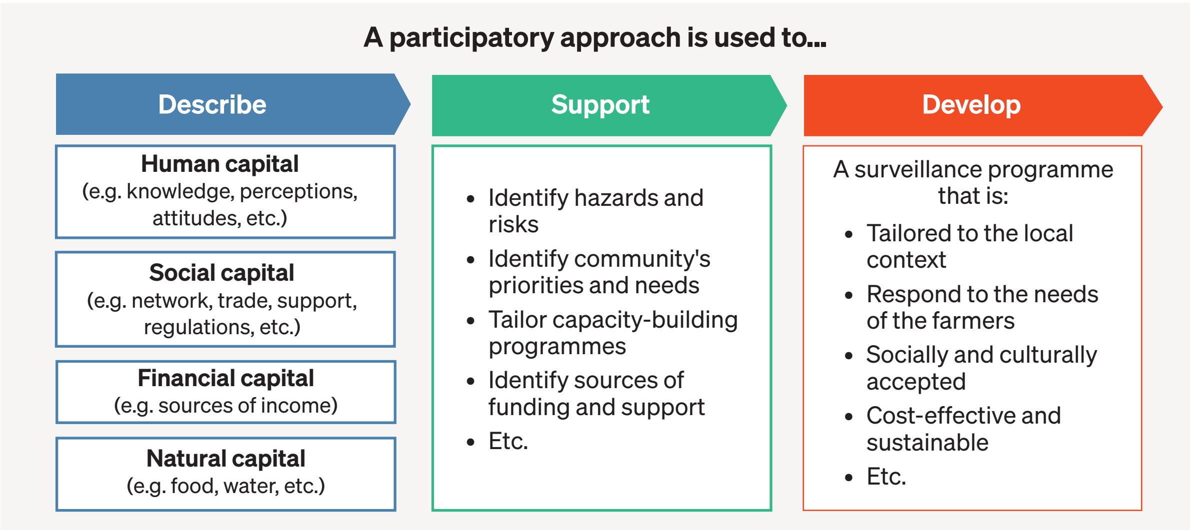

Figure 1. Key capitals of poultry systems described through a participatory approach to inform the design of socially accepted and sustainable disease surveillance programmes

2. Objective, target audience and framework of these guidelines

2.1. Objective

These guidelines provide practical recommendations to assist Veterinary Authorities (VA) and Veterinary Services (VS) in designing surveillance programmes for the detection of HPAIV in SHPS, tailored to the needs and contexts of resource-limited settings.

Such a surveillance programme should be integrated into the existing national surveillance system to ensure the early detection of HPAIV in poultry populations. This enables a rapid and effective response to control the spread of the virus and mitigate its impact.

2.2. Target audience

These guidelines are intended for national VA and VS involved in HPAI surveillance, as well as for wildlife, environmental and public health services, animal diagnostic laboratories and other One Health partners.

Implementation may benefit from collaboration with international and regional organisations, economic communities, the private sector, research institutions and civil society groups engaged in animal health and zoonotic disease prevention.

2.3. Framework

These guidelines are aligned with the *Global Strategy for the Prevention and Control of HPAI* by the Food and Agriculture Organization of the United Nations (FAO) and WOAH, which supports the strengthening of national surveillance systems to protect poultry value chains, safeguards livelihoods and promotes the health of humans, animals and ecosystems. This approach contributes to the United Nations Sustainable Development Goals 1, 2 and 3 (*No Poverty*, *Zero Hunger*, and *Good Health and Well-Being*).

The WOAH *Terrestrial Animal Health Code* (known as the *Terrestrial Code*) and the *Guide to Terrestrial Animal Health Surveillance* provide key reference frameworks for the design and evaluation of surveillance components and quality attributes. The Performance of Veterinary Services (PVS) Pathway offers tailored training and capacity-building tools, complemented by WOAH's competency and curriculum guidelines for veterinarians, VPPs and CAHWs.

These guidelines should also be used in conjunction with the WOAH Guidelines for Public–Private Partnerships in the veterinary domain, the *Manual of Diagnostic Tests and Vaccines for Terrestrial Animals* (known as the *Terrestrial Manual*) and the RISKSUR EVA tool (Surv-tool).

2.4. How to use these guidelines

Section 1 showcases the diversity and complexity of SHPS beyond the value chain and highlights the importance of community participation to address issues associated with poultry health and enhance effective disease surveillance programmes in such settings.

Section 2 outlines the main objectives, target audience and the international framework, standards and existing tools that link to this guidance.

Section 3 provides six practical phases for developing a surveillance programme in a participative manner — from defining objectives and mapping poultry systems to choosing methods, setting up communication and reporting pathways, and evaluating performance.

Section 4 focuses on training, including a Training of Trainers (ToT) model to strengthen local capacity.

The **Annex** contains ready-to-use tools, templates, figures and evaluation indicators to help users apply the guidance in the field.

3. Designing a surveillance programme for HPAI in small holder poultry systems

This section presents six key phases and considerations for designing a surveillance programme for HPAI in SHPS, particularly in resource-limited settings. A schematic summary of these phases can be found in **Figure 8** at the end of this section on page 16.

First, it is important to define the objectives and intended outcomes of the surveillance programme, ensuring alignment with the Member's national surveillance and policy framework for HPAI. Once these objectives are clear, the expected benefits, costs and resources should be carefully assessed across all stages of the surveillance programme – design, implementation and evaluation. Detailed resource allocation is especially important in resource-limited settings and should include identifying existing national capacities such as laboratories, local veterinarians, Veterinary Paraprofessionals (VPPs) and Community Animal Health Workers (CAHWs).

Although VPPs and CAHWs are not present in all countries, their contributions to providing basic animal health care and services to farmers and animal keepers is well recognised, particularly in remote locations and resource-limited settings. Members are therefore encouraged to involve these groups, where available, in the implementation of HPAI surveillance in SHPS. In addition, potential funding sources should be explored across both public and private sectors. Fostering public–private partnerships (PPPs) can enhance programme sustainability through shared resources and collaboration between the public sector and end-beneficiaries (e.g. poultry keepers) during the design of a surveillance programme, ultimately building trust between both sectors and leading to a better implementation (see WOAH's *Guidelines for PPPs in the veterinary domain*).

Phase 1 – Defining surveillance objectives

The objectives of the surveillance programme must be clearly defined, as they guide the design of surveillance activities, their duration, frequency and expected outcomes. These objectives also provide a framework for evaluating the programme's performance. Typical objectives include monitoring disease trends, supporting disease control measures, collecting data for animal or public health risk assessment and risk communication, justifying sanitary measures and providing trade reassurances.

Objectives should align with national strategies for preventing and controlling HPAIV and other infectious agents, while being adapted to local contexts and capacities. Depending on local circumstances, a steering committee for HPAI surveillance in SHPS may be established during programme design to oversee implementation and ensure effectiveness and sustainability (**Figure 2**).

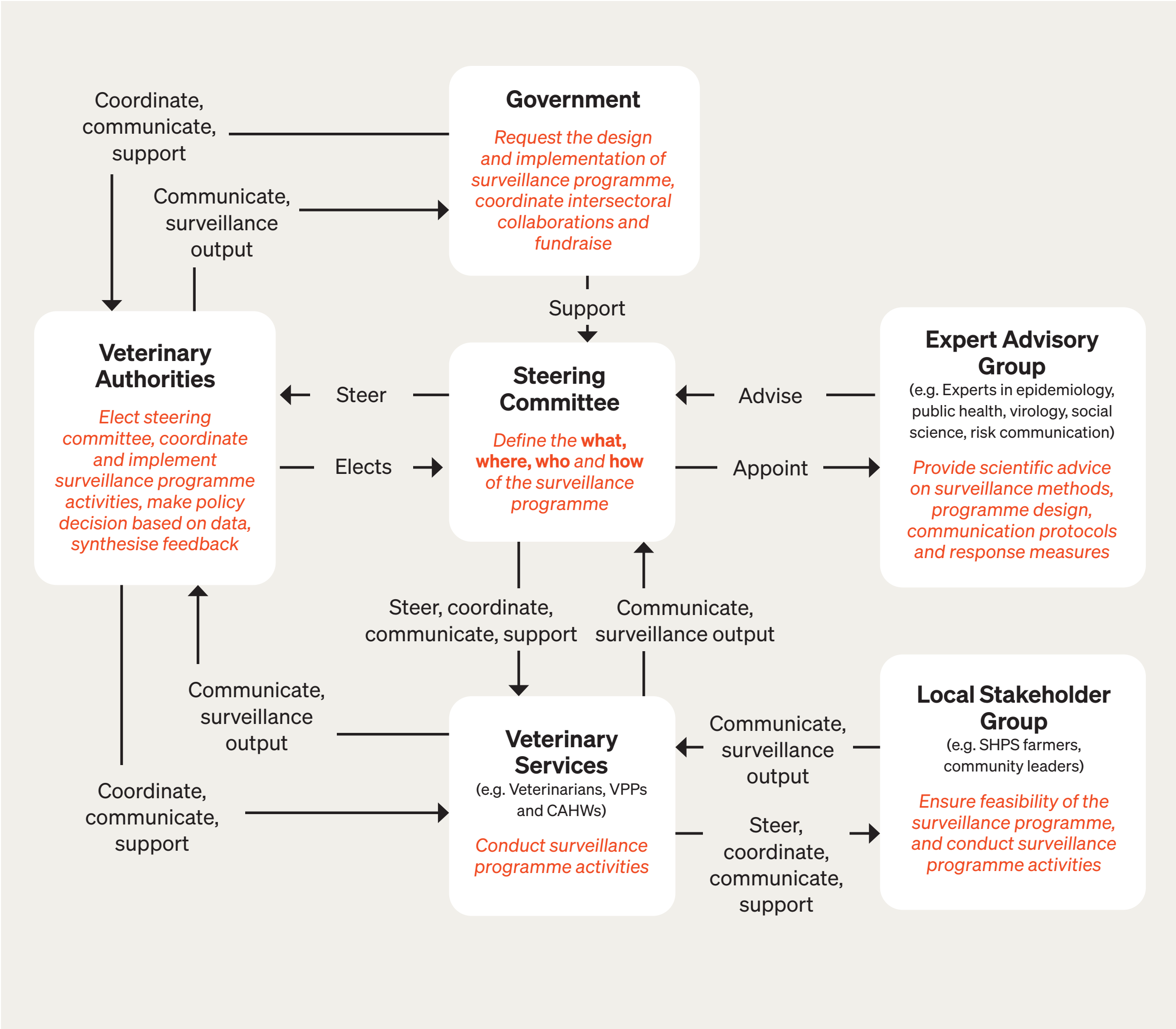

Figure 2. Example of a steering committee structure for the design and implementation of a surveillance programme

The composition and role of the steering committee will vary according to the local context, institutional structure and available resources

Phase 2 – Mapping the poultry production systems and value chain

Designing an effective surveillance programme for HPAI requires a thorough understanding of poultry production systems and the value chain – specifically, how stakeholders operate and manage poultry populations and products. Understanding the regular movement of birds, poultry products, materials, vehicles and people helps identify potential pathways for HPAIV spread within and between local poultry systems. This, in turn, informs strategic planning by focusing surveillance efforts where they will be most cost-effective.

Participatory mapping of local poultry production systems and value chains [19,22] helps identify key actors (e.g. farmers, traders, middlemen, vendors, slaughterhouse workers, retailers and consumers) and their epidemiologically relevant interactions. When combined with risk assessment, this information provides the foundation for designing an optimal surveillance programme [23]. Engaging SHPS farmers, local communities, poultry traders and other relevant stakeholders fosters a holistic understanding of existing veterinary and local knowledge, as well as experience from livestock keepers and extension workers [19], which can be explored further through a mapping workshop.

Mapping workshop setup

The objective of the mapping workshop is for VA, VS and stakeholders to jointly develop:

- **Spatial maps** showing the geographical layout of local poultry systems, and
- **Flow diagrams** illustrating the poultry value chain and related components that may influence the systems.

VA and VS should begin by identifying stakeholders involved in the poultry value chain, so that all major parts and themes of the local system are considered from the outset (see **Figure 3**).

Guiding questions for selecting participants (stakeholders):

- Who affects and/or is affected by the systems?
- Who has on-the-ground knowledge?
- Who has a strategic overview of the system?
- Who is often overlooked, and how can they be included?

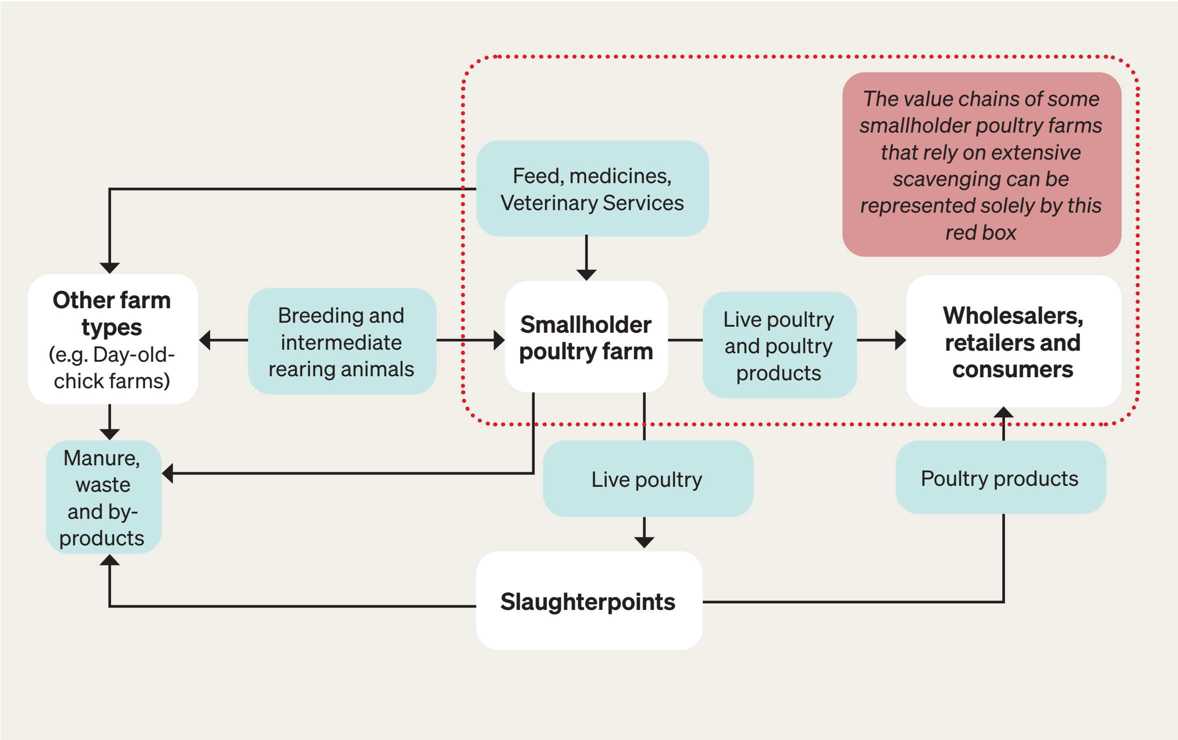

Figure 3. Example of a smallholder poultry value chain
The diagram illustrates the main actors and product flows in smallholder poultry systems. The structure and complexity of the value chain may vary depending on local conditions. In backyard or small-scale extensive scavenging production systems, feed sources often include kitchen scraps or natural foraging rather than commercial feed products, or new chicks may originate from natural incubation rather than purchased day-old chicks. Such systems have limited access to VS, and poultry products are typically consumed within the household

A mapping workshop should also provide an opportunity for VA, VS and local stakeholders to strengthen collaboration and build a shared understanding and ownership of the local poultry systems. The participatory process can elicit perspectives that are often marginalised and make them visible to more powerful actors (e.g. decision-makers), highlighting knowledge gaps within the production systems. To achieve this, the workshop design should be participatory, transparent and inclusive of diverse stakeholders and viewpoints (**Box 2**).

BOX 2.

STRATEGIES FOR INCLUSIVE STAKEHOLDER ENGAGEMENT DURING THE MAPPING WORKSHOP

- Keep workshop size small while ensuring representation across the SHPS value chain and other poultry production systems (e.g. small-to-large commercial poultry producers).
- Make the process transparent and open. Announce workshop in advance, share output in real-time during the session, and welcome contributions from all interested participants.
- Iteratively expand participation. Continuously ask stakeholders to identify missing voices, especially those bridging different poultry production systems, before and during the workshop.
- Create a positive and creative environment that encourage trust, strengthens relationships and inspires ongoing support for the next steps.

Suggested key questions and steps for the mapping workshop are provided in **Box 3** and **Figure 4**, respectively. These should not be viewed as fixed instructions but rather adapted to the scale and specific context of the poultry production systems. It is recommended to conduct mapping workshops in several representative locations, so that findings can be consolidated at the national level to inform surveillance programme design.

For additional guidance on participatory mapping and on how value chain mapping can support risk assessment and the design of surveillance strategies, please refer to Ameri *et al.* [19] and FAO [23], respectively.

BOX 3.

KEY QUESTIONS TO ADDRESS DURING THE MAPPING WORKSHOP

The purpose of these questions is to improve understanding of local poultry production systems and their components. They help identify risk factors associated with the incursion and spread of HPAIV, as well as opportunities and challenges for designing and implementing a sustainable and effective surveillance programme for HPAI in SHPS at community or regional levels. These questions should be adapted to the scale and specific context of each poultry production system.

1. How many different types of SHPS exist in the area?
2. What types and species of poultry are raised in these SHPS (e.g. commercial or purebred chickens, mixed-breed chickens, ducks, guineafowl, quail, pigeons, etc.)?
3. What are the husbandry and ownership characteristics of these SHPS?
4. How are SHPS distributed within the community or region?
5. How are SHPS connected? Describe the components within and outside the value chain, and how they interact over time and space.
6. What are the key routes, people and organisations involved in the value chain? Describe how they influence the value chain and the dynamics of the overall poultry production systems.
7. What are the common practices, rewards, cultural preferences and training of those involved in the value chain?
8. What factors contribute to HPAIV transmission in SHPS and the value chain? How can these risks be mitigated? What political, economic, socio-economic, technological, environmental and legal factors are related to risk management?
9. How are SHPS and stakeholders situated in relation to nearby large bodies of water or areas that overlap with migratory bird routes?
10. Are there any non-trade transaction connections in the value chain that pose a risk for HPAIV transmission (e.g. fighting cock practices, gifting birds to visitors)?

A suggested 4-step process for mapping local poultry production systems and value chains

STEP 1 – Establish the boundary of the local poultry production system

- This step ensures the mapping process remains tractable.
- Get consensus from stakeholders on the extent of the system (or the geographical scope of the map). This could be the surrounding area of a smallholder poultry farm, a village, a community, a district or a province.

STEP 2 – Map key features of the local poultry production system

- Draw the key features of the local poultry system and value chain on a map, such as:
  - farm locations (SHPS, intensive poultry farms and other farm types)
  - live bird markets, slaughtering points, processing plants
  - Veterinary Services, carcass disposal sites
  - main roads, large water bodies (e.g. lakes), etc.

STEP 3 – Create a flow diagram of the local poultry value chain

- Describe the movement of the birds, poultry-related products, materials (e.g. feed), vehicles and people in the value chain based on the knowledge of activities and movement patterns of stakeholders (e.g. common trade routes).
- Draw and label boxes to represent stakeholders and sites. Use arrows between boxes to indicate flows of poultry, poultry-related products, materials and other production inputs and outputs between entities.

STEP 4 – Conduct a preliminary qualitative risk assessment

- Consider all possible transmission pathways of HPAIV and identify opportunities for spread in the described value chain. Identify risks and hotspots throughout the value chain:
  - Can a disease agent enter here? (source, route)
  - Can a disease agent survive here? (conditions, treatments)
  - Would a disease agent be noticed here? (surveillance)
  - Can a disease agent be carried out from here? (destination, route)

Mapping output to inform and guide the design of the surveillance programme

Figure 4. Suggested steps for conducting a mapping workshop

The steps outlined should be adapted to the scale and specific context of poultry production systems

The mapping output should inform the design of the surveillance programme, including the selection of surveillance types and methods (**Phase 3**) and sampling units (**Phase 4**). For example, mapping may reveal that some smallholder poultry farms are epidemiologically isolated or not connected to the broader local poultry system. In such case, it is important to document these findings, as they help determine whether these farms should be excluded from the study population and justify how well the study population represents the target population (**Phase 4**).

Furthermore, the mapping output can enhance understanding of local poultry stakeholders, including their operations, profitability, opportunities, constraints, and their knowledge and perceptions of HPAI prevention and control. This understanding can help identify factors that influence reporting behaviour and engagement in a given local context (**Phase 5**).

Phase 3 – Selecting surveillance types and methods

The surveillance types and methods selected for the programme should align with the purposes and objectives outlined in **Phase 1**. A combination of surveillance types (e.g. active or passive), methods (e.g. risk-based surveillance, participatory disease surveillance) and activities (e.g. routine sampling at slaughterhouses) may be required to complement one another.

When selecting surveillance types and methods, the following criteria should be considered:

- The intended target population (**Phase 4** – Defining the population under surveillance)
- Feasibility of sampling the population, considering potential restrictions (e.g. geographical barriers) (**Phase 4** – Defining the population under surveillance)
- The data required, including data sources and plans for processing and analysis
- Desired levels of surveillance performance (e.g. expected sensitivity and timeliness)
- Available financial resources
- Personnel requirements, including the necessary technical and managerial skills
- Laboratory infrastructure and capacity
- Governance structure
- Regulatory framework
- Incentives and disincentives for engaging key actors

In the Annex at the end of this publication, you can find further information to help with these considerations. **Table A1.1** provides a list of surveillance methods that may be applied when designing an HPAI surveillance programme for SHPS. It provides guidance on when to use each method, outlines their relative advantages and limitations, and offers suggestions for effective implementation in resource-limited settings. **Table A1.2** presents a template to help Members select and tailor surveillance methods for HPAI in SHPS for their own unique context.

In resource-limited settings, PDS as an active clinical surveillance method is recommended, as it is supported by laboratory diagnosis confirmation (**Figure 1**). Additional guidance on surveillance types and methods can be found in the WOAH *Guide to Terrestrial Animal Health Surveillance*.

Once surveillance methods are selected, the associated diagnostic procedures must be clearly defined, as detailed below.

Case definitions:

Case definitions may include clinical signs, necropsy findings, laboratory test results and classifications based on the certainty of results gained from the different diagnostic tools used (e.g. *suspected*, *probable* or *confirmed*). The case definition should be reviewed regularly to reflect changes in programme objectives, diagnostic advancements and evolving understanding of the disease. Implementation of prevention and control measures of HPAI, such as vaccination, can also impact the case definition. It is essential that all stakeholders involved in the surveillance programme agree upon and clearly understand the adopted case definition.

Monitoring poultry for clinical evidence of HPAI:

HPAIVs are generally highly transmissible and can cause rapid and severe mortality of entire flocks. The detection of a sudden increase in bird mortality may indicate a *suspected case*, which can later be labelled as a confirmed case with laboratory diagnosis (**Box 4** and **Figure 5**).

When establishing reliable reporting thresholds for a *suspected case*, several factors should be considered, including poultry species (e.g. chicken, duck), stocking size and density, genetics, immunity status (e.g. vaccinated or non-vaccinated), age and the presence of other pathogens. Additional considerations for HPAI case detection include whether other diseases are present with similar clinical signs and that exhibit high mortality (e.g. velogenic viscerotropic Newcastle disease). Where vaccines for these diseases are in use, they can enhance confidence in detecting HPAI [24].

However, the co-circulation of low pathogenicity avian influenza viruses can complicate the molecular and serological diagnostic surveillance in both vaccinated and non-vaccinated flocks affected by HPAI [25]. Similarly, the use of DIVA (Differentiation of Infected from Vaccinated Animals) vaccines may negatively affect the specificity of serological tests used to measure immune responses [26]. In contrast, reverse transcription quantitative polymerase chain reaction (RT-qPCR) testing is considered more reliable for surveillance in vaccinated flocks due to its high sensitivity and specificity. Yet, its use may not be practical in resource-limited settings due to its cost and the level of expertise required.

BOX 4.

CASE STUDY: SUSPECTED AND CONFIRMED CASE DEFINITION USED IN A PARTICIPATORY DISEASE SURVEILLANCE (PDS) PROGRAMME FOR HPAI IN YOGYAKARTA PROVINCE, INDONESIA, 2008 [27]

The government of Indonesia, in partnership with FAO, implemented a PDS programme for detecting HPAI outbreaks in household chicken flocks. The programme utilised a set of clinical case definitions complemented by commercial rapid antigen test kits. Cases matching the clinical case definition but without rapid test results were labelled as *suspected cases*.

Following positive rapid test results, these cases were reclassified as *confirmed cases*. Considerations were made to vary clinical case definitions based on the scale of the household chicken flock (small backyard flocks *versus* large commercial units) and the flock's vaccination status (vaccinated *versus* non-vaccinated). For example, the sudden death of one or more birds in a backyard indicated a suspected case, while in large commercial units (hundreds to thousands of birds), a suspected case was defined by unexplained mortality exceeding 1%.

Testing:

When a suspected case of HPAI is identified, point-of-care screening tests can be used to support preliminary diagnosis. In resource-limited settings, where VS may not be present in rural or remote areas, rapid test kits can be distributed to trained local stakeholders (e.g. CAHWs, poultry vaccinators, poultry traders and/or poultry producers) under the supervision of the VA and VS. With appropriate training, these stakeholders can act as intermediaries to perform testing and relay results.

However, in many resource-limited settings, distributing rapid screening tests directly to local farmers may not be feasible due to the costs, training needs and logistical challenges involved. In such cases, training and integrating CAHWs into the VS system to perform rapid testing for HPAI may be more effective. At a minimum, an awareness and reporting mechanism should be established to enable field diagnosis by VS.

Currently, there are several commercially available solid-phase antigen-capture enzyme-linked immunosorbent assay (AC-ELISA) kits that can detect influenza A virus in clinical specimens. Only kits validated for veterinary use in accordance with the recommendations outlined in the *Terrestrial Manual* should be used. As with all rapid screening tests, false positive and false negative results may occur. The choice of rapid screening test should therefore depend on the surveillance objectives, available resources and feasibility of implementation, as well as the circumstances of the surveillance activity at a given point in time. For example, during the early stages of a control programme, a low-cost rapid screening test with good sensitivity but moderate specificity may be acceptable when the disease prevalence is high. As disease prevalence decreases during the course of the control programme, the same test may generate a higher proportion of false positives, thus increasing the uncertainty of the results. The brand names and specifications of all rapid test kits used should be documented in the surveillance database. Other point-of-care tests, such as portable polymerase chain reaction (PCR) tests, may also be considered.

A suspected case with a positive screening test is classified as a *probable case* (**Figure 5**). To identify a *confirmed case*, laboratory-based confirmatory tests – such as virus isolation, haemagglutination inhibition, and PCR – should always be conducted on all suspected cases by national or reference laboratories. The full list of recommended confirmatory laboratory tests for HPAI is available in the WOAH *Terrestrial Manual*.

However, confirmatory laboratory tests can be costly, time-consuming and technically demanding, which may be unfeasible in resource-limited settings. The selected confirmatory tests must therefore align with existing national diagnostic capacity, and ensuring access to testing facilities is vital for the surveillance programme's success. Where national or regional laboratory capacity is insufficient, Members may consider sending specimens to a WOAH Reference Laboratory.

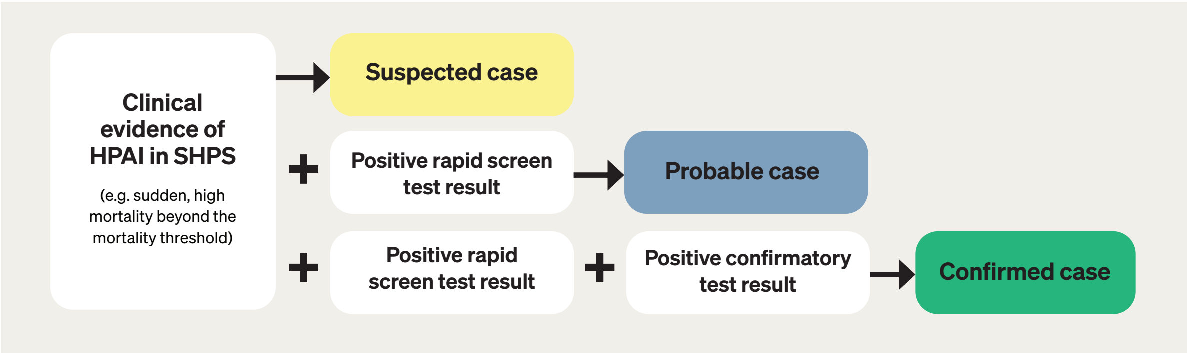

Figure 5. Suggested diagnostic procedures for defining a suspected, probable and confirmed HPAI case

All considerations related to sampling should be clearly defined during the design of the surveillance programme. These include the use of appropriate personal protective equipment (PPE), procedures for sample collection, selection of sampling equipment, and the correct packaging, storage and transport of collected samples. The programme should also specify all actions to be taken when suspected or confirmed cases are detected. This includes investigation procedures (and the use of any screening or confirmatory tests), reporting to officials and control measures (**Box 5** below and **Table A2** in the Annex), which should also be specified in the surveillance programme.

BOX 5.

CONSIDERATION OF RESPONSE MEASURES FOR SUSPECTED, PROBABLE OR CONFIRMED CASES OF HPAI

- Response measures should align with both the national and local strategies for controlling HPAI, as surveillance in SHPS is a key component of the country's overall surveillance and response framework for HPAI.
- Response measures should be practical and feasible, ideally developed in collaboration with farmers, local stakeholders, VS and VA. These measures should be clearly defined based on the level of certainty regarding the case (i.e. suspected, probable or confirmed cases of HPAI). This may include restricting movement or implementing biocontainment practices tailored to the local context.
- It is also crucial to provide training on the safe disposal of sick birds to prevent human or animal access to and consumption of infected birds. Additionally, long-term training should focus on basic personal hygiene procedures, including handling dead birds, culling and the proper disposal of carcasses.
- Veterinary Authorities should leverage and seek assistance from VPPs and CAHWs whenever possible, particularly in remote locations.

When implementing the surveillance programme, diagnostic procedures must be regularly evaluated to detect any misdiagnoses related to sudden deaths or increased mortality. Tools such as **Table A3** (see Annex) can assist in this process. In addition to assessing the performance of each diagnostic tool, it is essential to review the resources required, as these factors affect the cost-effectiveness of the diagnostic process and guide decisions on whether methods should be continued or revised. Identifying the causes of misdiagnoses is also critical; if human error is involved, adjustments to training and supervision may be necessary. Members applying these guidelines should evaluate their existing infrastructure and resources to ensure that their national surveillance systems are optimised for HPAI detection.

Phase 4 – Defining the population under surveillance

The selection of target and study populations is guided by the surveillance objectives and available resources, and both must be clearly defined in the surveillance programme. The *target population* refers to the broader population at risk – the group about which inferences will be drawn from the surveillance data. The *study population* consists of the individuals or units from which information or samples are collected; it is typically (though not always) a subset of the target population and should be defined and justified in the surveillance programme.

Depending on the scale of the surveillance programme (i.e. national, regional or local), resource constraints and the willingness of farmers to participate, the target and study populations may or may not be the same (**Figure 6**). Selecting an appropriate study population ensures that surveillance findings can be reliably generalised to the target population. A risk-based approach may be used, taking relevant epidemiological risk factors into account.

Additionally, the surveillance programme should specify the administrative units covered (e.g. farm, village, zone, province) and the epidemiological unit of observation (e.g. individual birds, flocks or villages in extensive production systems) (see **Box 6**). Additional guidance on defining target and study population for surveillance programmes can be found in the WOAH *Guide to Terrestrial Animal Health Surveillance*.

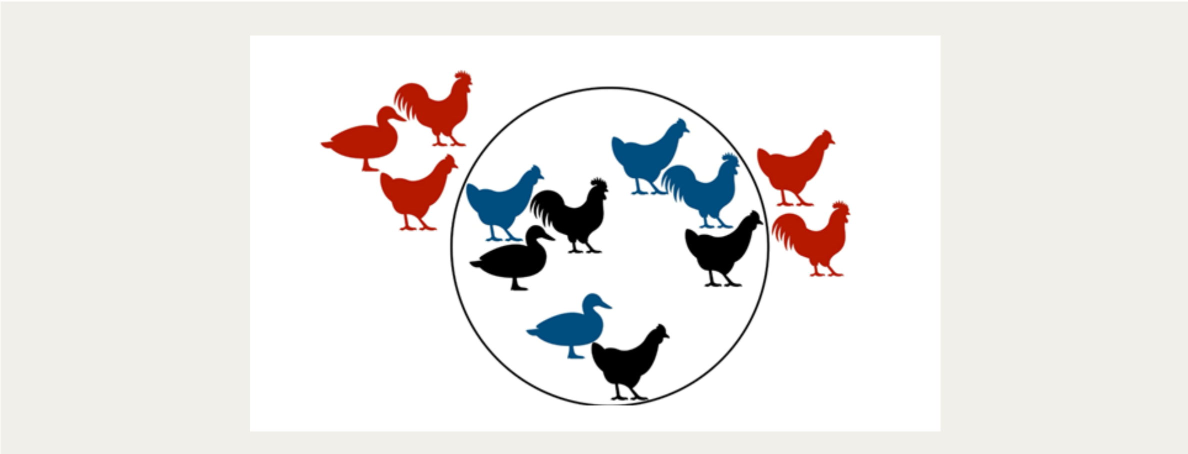

Figure 6. Example of the relationship between target and study populations in a surveillance programme

The target population (all birds shown in black and blue within the circle) represents the poultry at risk included in the surveillance objective. The study population (birds shown in blue within the circle) refers to the subset actually sampled. Birds shown in red, outside the circle, are excluded from the programme (e.g. large commercial poultry production is not part of the target population). Depending on the programme's scale (e.g. national, regional or local), available resources and farmer participation, the study population may fully or only partially overlap with the whole target population. If there is a full overlap, study and target samples are considered the same

BOX 6.

SUGGESTED INFORMATION TO BE DESCRIBED WHEN DEFINING THE TARGET AND STUDY POPULATION UNDER SURVEILLANCE

Target population (general population at risk):

- Target population size (e.g. number of birds, flocks, or farm per community, zone or region)

Study population (subpopulation of target population):

- Estimated study population size for the targeted populations (e.g. number of birds, flocks or farms per community, zone or region)
- Specific HPAI variables for the targeted population

Unit of observation:

- E.g. individual birds, flocks, farms
- Geographic or other spatial measures (e.g. flock or farm per km², proximity to bodies of water or other geographic features of importance)

Administrative or political unit:

- E.g. farm, neighbourhood, village community, zone, region

Animal information:

- Species, breed
- Age, sex
- Production phase
- Husbandry practices or management (e.g. small-scale extensive scavenging *versus* small-scale commercial farms)

Risk factors

- Population risk factors that may influence interpretation of surveillance data (e.g. confounders/effect modifiers)

Phase 5 – Defining communication and reporting protocols

The success of a surveillance programme depends on efficient information flow among stakeholders through clearly defined communication pathways (**Figure 7**). The communication plan should also specify how information is reported and shared, including the methods, frequency and responsible parties.

As the first point of detection and likely the group most affected by HPAIV-positive cases, farmers should be primary beneficiaries of the surveillance system. To ensure engagement and promote timely reporting, investigation results must be promptly communicated to farmers, along with recommendations for response measures (**regardless of whether the disease is confirmed as HPAI or another condition**).

The communication plan and the methods for disseminating information must comply with the Member's national regulations on data privacy and confidentiality. It should be defined during the surveillance programme design phase, with all relevant stakeholders trained in reporting procedures, information recording protocols and privacy protection measures. **Maintaining participant privacy and anonymity is critical to prevent social stigmatisation and the consequent disincentive to participate and report**.

It is also important to identify factors that contribute to delayed reporting or underreporting. While awareness of HPAI can increase readiness to report outbreaks [28], education alone does not guarantee better reporting [29].

Reporting behaviour is influenced by multiple factors, including:

- Uncertainty about HPAI clinical signs and reportable situations
- Inadequate awareness of reporting, processing and response procedures
- Fear of social and economic repercussions (from both true and false positive reports)
- Negative perceptions of response effectiveness
- Mistrust and dissatisfaction with animal health authorities due to past experiences
- Lack of financial or non-financial incentives to report

Understanding these factors requires consideration of farmers' knowledge, experience, beliefs and local social norms. Insights gathered from Phase 2 (Mapping the poultry production systems and value chain) can help inform this process.

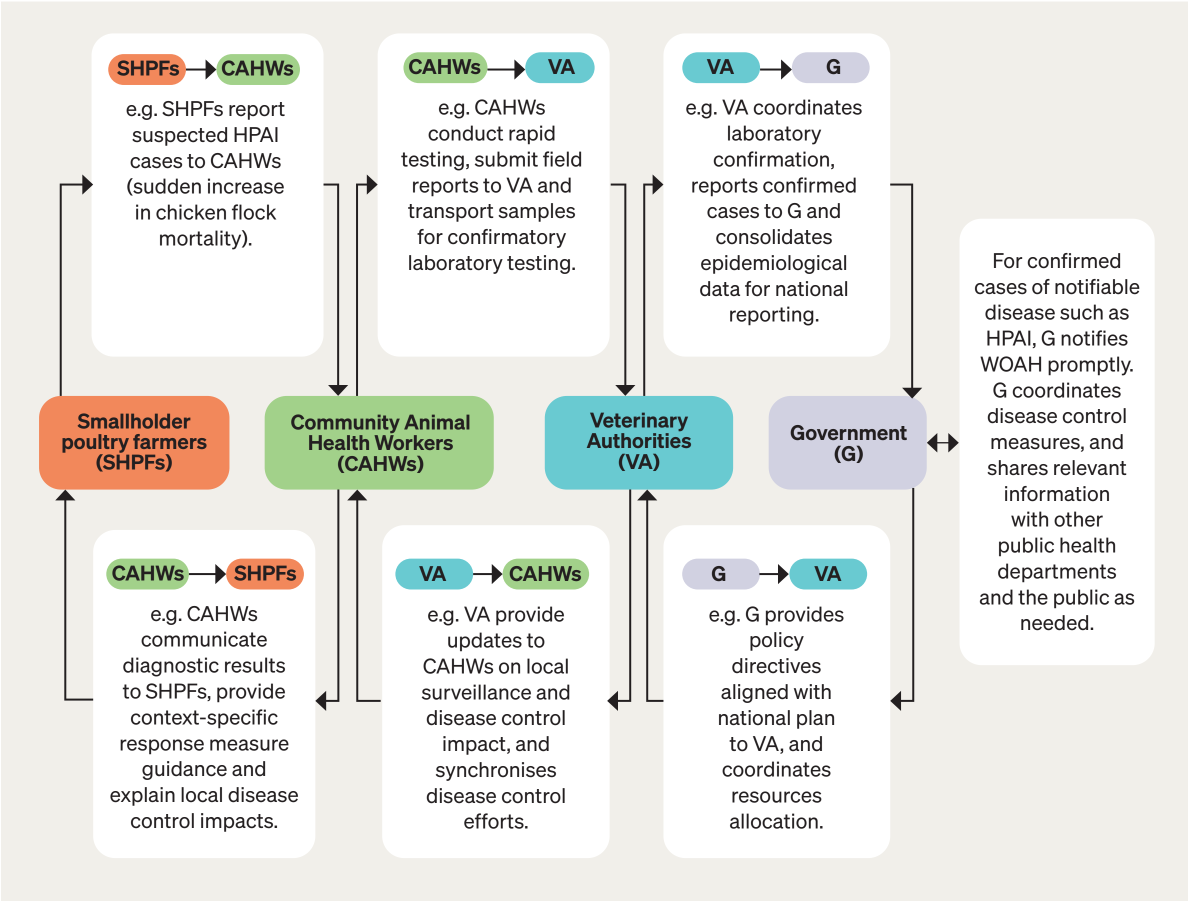

Figure 7. Example of communication and information flow during a surveillance programme

The stakeholders involved, the type of information exchanged and the level of detail will vary depending on the local context and should be adapted accordingly

Phase 6 – Assessing and evaluating the surveillance programme

Surveillance programmes should be routinely evaluated to ensure they meet their defined objectives and to identify opportunities for improving quality, efficiency and usefulness. Evaluation activities should be prioritised, adequately resourced and carried out systematically, transparently and reproducibly by qualified assessors.

Table A4 in the Annex provides a suggested list of surveillance evaluation indicators for HPAI in SHPS in resource-limited settings. These indicators should be adapted to each Member's specific context and updated regularly. Traditional surveillance system attributes, such as simplicity, flexibility and acceptability, also remain important considerations for SHPS in resource-limited settings.

This list of surveillance evaluation indicators (**Table A4**) should be used in conjunction with the WOAH *Guide to Terrestrial Animal Health Surveillance* and the RISKSUR EVA tool (Surv-tool). It is also important to assess how animal welfare principles can be integrated throughout all stages of designing and implementing the surveillance programme. Members may follow their own animal welfare regulations or, where these are lacking, apply the international standards outlined in the WOAH *Terrestrial Code*.

The ultimate goal of the evaluation process is to produce practical solutions. Findings should be translated into actionable plans that enhance the surveillance programme's effectiveness and support the achievement of its overall surveillance objectives.

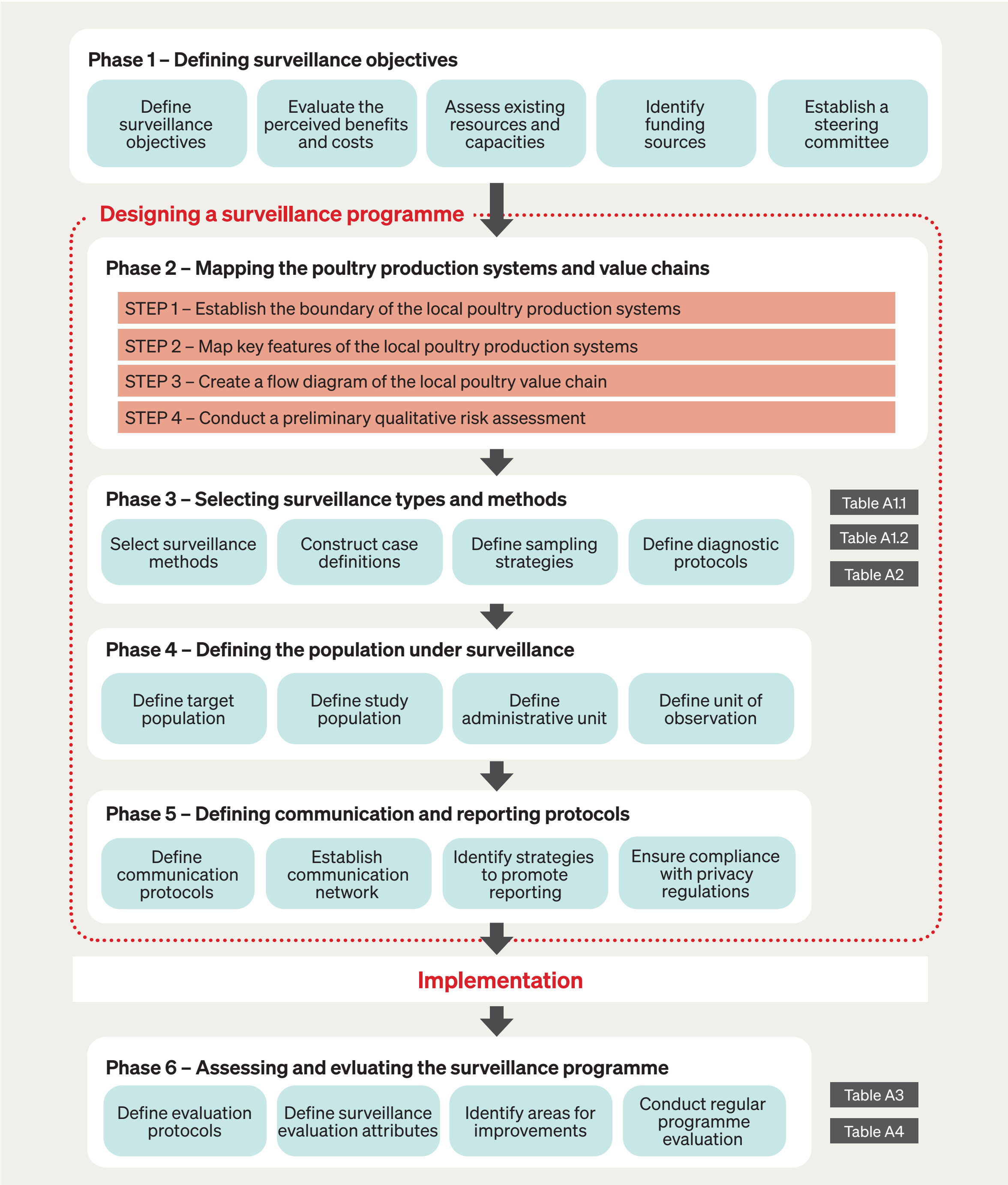

Figure 8. Suggested phases for designing, implementing and evaluating a surveillance programme for HPAI in SHPS within resource-limited settings

Dark grey boxes indicate the relevant tools provided in the Annex for each phase. The phases are intended as guidance and should be adapted to the scale of the poultry production systems and their unique local context

4. Training programme

4.1. Scope and approach

The approach described in these guidelines requires a sound understanding of the concepts and tools related to both disease surveillance and participatory methods. It also depends on the active engagement of diverse stakeholders from different socio-cultural backgrounds and professional disciplines to enable knowledge exchange and build a shared understanding of the complexity of local SHPS beyond the value chain.

Awareness campaigns highlighting intersectoral collaboration to safeguard domestic animal health and welfare, and food sovereignty should be designed to motivate farmer participation and support the selection of training participants who will engage in the surveillance activities (hereafter referred to as the task force). Government agencies may also consider linking compulsory participation in surveillance activities to eligibility for specific community benefits or support schemes (in any form).

Given the complexity and diversity of SHPS, and the importance of this surveillance programme as part of a larger (i.e. national) surveillance system, the training programme must be tailored and participatory in nature. It should integrate the different skills and expertise of stakeholders to ensure collective success (**Box 7**). Understanding the priorities of farmers and communities, coupled with effective communication, is essential for fostering long-term engagement and ensuring the sustainability of the surveillance programme.

Although the following training programme can be implemented as a standalone initiative, it is recommended that it complement existing frameworks such as the Performance of Veterinary Services Pathway, the *Handbook for Planning and Managing Community Animal Health Workers (CAHWs) Programmes* and the *Avian Flu School International Course Guide*.

BOX 7.

KEY CONSIDERATIONS FOR DESIGNING AN EFFECTIVE PARTICIPATORY TRAINING PROGRAMME

- Keep the group size manageable and ensure that all relevant sectors are adequately represented.
- Foster a positive and inclusive atmosphere throughout the workshops.
- Maintain transparency by keeping participants informed about progress, outcomes and next steps.
- Use plain, accessible language when explaining epidemiological or technical concepts.
- When working with local and indigenous groups, consider engaging a translator if necessary to ensure clear communication.
- Identify key terms that require translation or local interpretation and develop a shared glossary with participants.
- Respect participants' time by agreeing on the number and duration of sessions in advance and adhering to the schedule.
- For sessions lasting longer than three hours, plan short breaks every hour. Coordinate with the community to provide refreshments where possible.
- Consider offering a stipend or other form of compensation to acknowledge participants' time and contributions.
- Where farmers or communities receive government support, participation in the programme may be linked to eligibility for such benefits.
- Adapt the programme design to the socio-cultural context of the community or region to maximise engagement, acceptance and long-term success.

Source: Vétérinaires Sans Frontières International [30]

4.2. Training of trainers (ToT) model

The Training of Trainers (ToT) model (**Box 8**) equips VA, experienced field veterinarians and veterinary epidemiologists (master trainers) to train and supervise local animal health workers (trainer participants; e.g. local veterinarians, VPPs and CAHWs). The ToT model aims not only to ensure the efficient and effective transfer of essential skills needed for implementing the surveillance programme but also to strengthen the overall health and welfare of farmed animals in the community. Its success depends on the active involvement of experts in participatory methods from various disciplines (e.g. social scientists and educators) to facilitate training and support capacity development (**Figure 9**).

BOX 8.

BRIEF DEFINITION OF THE TRAINING OF TRAINERS (TOT) MODEL

ToT is an empirical model that supports the effective transfer of skills, providing a clear pathway for trainees to quickly gain knowledge and acquire the necessary skills to perform their jobs in a constantly changing environment.

Source: Moton B. and Daniels C. [31]

Figure 9. Training of Trainers model

Based on the U.S. Centers for Disease Control Training and Professional Development in Schools [32]

4.3. Key participants in the Training of Trainers model

Given the diversity of stakeholders involved (both directly and indirectly) and the technical nature of disease surveillance activities, it is crucial that the task force receives comprehensive yet accessible training. The success of the ToT model relies on selecting the right participants for each of the two training stages described in these guidelines. Ideal candidates should possess not only knowledge and skills related to the surveillance programme but also a genuine commitment to improving the overall health and welfare of domestic animals in their communities (**Figure 10**).

Identifying suitable candidates enables the design of an effective training programme, helps define clear participant roles and fosters ongoing community engagement. Participants selected for the surveillance programme's task force training must meet specific criteria that are established in consultation with the community. Key considerations for selection are outlined in **Box 9**.

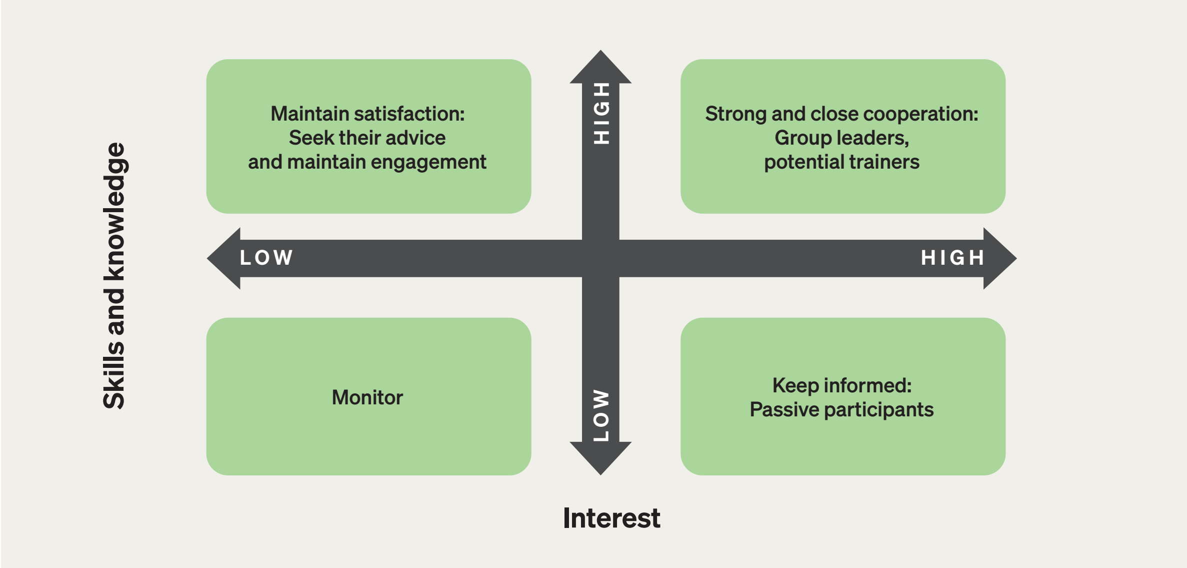

Figure 10. Relevance of stakeholders involved in the surveillance programme, based on their interest, desired skills and knowledge
Adapted from Styk and Bogacz [33]

BOX 9.

CONSIDERATIONS FOR THE SELECTION OF THE TARGET AUDIENCE (TRAINEES)

Who should be considered for the training?

- Selection criteria should align with the community's customs, values and traditions.
- Inclusivity matters: building a diverse team – including youth, women, elders and members of vulnerable groups – can greatly enhance the success of the surveillance programme. However, it is essential to respect and adapt to local customs.
- Participants should be well-known and respected within the community to foster trust and encourage acceptance of the programme.
- A basic level of literacy (reading and writing) is recommended.
- Participants should have sufficient time available outside their family and community duties to engage in the training. Compensation – financial or in-kind – may be provided, as agreed by all stakeholders.
- Most participants should have in-depth knowledge of community dynamics and local farming practices, as they will play a key role in the design and implementation of the surveillance programme.

Source: Vétérinaires Sans Frontières International [30]

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4.4. Training programme content

The training programme is structured into two main stages, with the effectiveness of the second stage depending on the success of the first.

4.4.1. Training on participatory tools and methodologies to design a PDS programme

- **Target audience (trainees)**: Local and regional VA and VS involved or expected to be involved in the design of the surveillance programme. Trainees should have prior experience and knowledge of the regions and communities where the programme will be implemented, as well as familiarity with disease surveillance design and implementation.
- **Facilitators (trainers)**: Professionals with expertise in designing participatory workshops in resource-limited settings, ideally with knowledge of the regional context and communities where the activities will be implemented.

- **Expected outcomes:**
  - Local VS are equipped with the skills and knowledge to engage with farmers and community representatives in describing the local poultry production systems, both within and beyond the value chain, including the socio-economic and cultural components that define the system and its dynamics.
  - Local VS can apply participatory techniques to engage farmers, collaborate with community representatives and/or NGO partners, as well as design and facilitate the implementation of surveillance programmes that align with national systems while respecting local socio-cultural contexts.

- **Workshop flow:**
  - Encourage participants (VS) to share experiences of engaging with smallholder poultry farmers and to discuss the region's social and economic context, identifying both strengths and limitations. This will help facilitators select appropriate tools to improve community engagement.
  - Once the regional context has been described and experiences shared, facilitators should lead the conversation to the following topics (adapted from Hovmand [34]):
    - Define criteria for identifying members of the task force (i.e. stakeholders to be trained for the surveillance programme).
    - Identify relevant stakeholders and clarify their potential roles in enhancing engagement and performance of the surveillance programme. This should include the private sector, government and non-government agencies, and any other sector that can aid the success of the programme.
  - Present methodologies for participatory workshops and mapping of local poultry production systems, using case studies and examples drawn from the target community for exercises.
  - Develop a shared terminology and effective communication approach appropriate to the cultural and linguistic context. Identify keywords for translation into local languages (particularly in areas where indigenous languages are prevalent). Engage a local translator where possible.
  - Draft a training agenda and plan for the second stage (see **Section 4.4.2**) and conduct a pilot training programme.
  - Evaluate the results of the pilot training programme; identify and refine the training as needed.
  - Report results and seek approval from relevant authorities to implement the second stage of the training programme.

4.4.2. Training on the implementation of the PDS programme and related activities

- **Target audience (trainees)**: Farmers, local VS, community representatives and other actors identified as supporting the implementation of the surveillance programme.

- **Facilitators (trainers)**: Professionals with experience in designing participatory workshops in resource-limited settings, as well as VA and VS staff previously trained in participatory methods (see **Section 4.4.1**) and surveillance practices.

- **Expected outcomes:**
  - Trainees are able to describe and visualise the local poultry production systems, within and beyond the value chain, and their socio-cultural and economic context.
  - Trainees can identify potential risk pathways for the introduction and spread of HPAI, through (and beyond) the poultry value chain. This includes interactions with stakeholders or environmental components of the community system that are not part of the value chain but interact with its actors (e.g. wild birds sharing the ecosystem with free-ranging poultry flocks; neighbours with no poultry but occasional connection with the farmers and outside markets).
  - Trainees can co-design a community-tailored surveillance programme that complements national or regional HPAI surveillance systems.
  - Trainees are capable of identifying and reporting suspected HPAI cases within their flocks and communities, and of implementing effective response actions to prevent further spread.
  - Trainees are prepared to respond appropriately to disease events, regardless of HPAIV confirmation status.

- **Workshop flow:**
  - Outline the HPAI surveillance programme's objectives and its relevance at local and regional levels.
  - Use participatory workshops to map local poultry production systems, identifying key components and their interactions within and beyond the value chain.
  - Identify possible risk pathways for HPAI introduction and spread.
  - Assess available resources and requirements to design a feasible strategy.
  - Select a community task force (e.g. farmers, CAHWs and community members) to implement the surveillance programme in the community.
  - Train the community task force on diagnostic tools and reporting procedures as defined in the surveillance programme (**Section 3**). This includes:
    - Case definitions
    - Relevant data collection (e.g. vaccination history for HPAI and other diseases)
    - Diagnostic tests
    - Report and response procedures
    - Safe carcass disposal and isolation of suspected cases
    - Response actions for HPAIV-negative cases
    - Use of evaluation tools
  - Rehearse the surveillance procedures and refine the programme based on participant feedback.

4.5. Evaluating and revising the training programme

The efficacy of the training programme should be assessed through feedback from both trainers and trainees, as well as by monitoring the performance of trainees over time. VS and training partners should work together to identify any gaps in knowledge, skills or understanding that emerge during the participatory process. Regular refresher training should be planned and tailored to address identified needs.

The following criteria and actions are suggested for evaluating the success of the training programme:

- **Evaluation criteria**
  - Knowledge retention: assess how well trainees retain information related to the PDS programme and disease management practices.
  - Engagement and participation: evaluate the level of stakeholder engagement during workshops and training sessions.
  - Community acceptance: measure the community's trust and acceptance of the surveillance programme.
  - Practical skills: assess the ability of trainees to apply acquired knowledge skills in real-world settings.
  - Feedback and adaptation: evaluate how effectively feedback is incorporated into the revisions or updates of the training programme.
- **Evaluation actions**
  - Surveys and feedback forms: collect structured surveys from trainers and trainees to gather feedback on the training programme's effectiveness and areas for improvement.
  - Observational assessments: conduct direct observation during workshops to evaluate engagement and participation.
  - Performance metrics tracking: track key performance indicators over time (e.g. number of suspected cases reported, response times).
  - Community consultations: organise community meetings to gauge acceptance and gather feedback on the training programme from a broader audience.
  - Programme revision workshops: host workshops to revise and refine the training programme based on collected feedback and performance data.

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Annex

Table A1.1. Surveillance types and methods for consideration when designing a surveillance programme for High Pathogenicity Avian Influenza (HPAI) in smallholder poultry systems (SHPS) in resource-limited settings

The selection of surveillance types and methods should align with the established objectives of the programme and take into account the specific local context of each country. For additional guidance on surveillance types and methods, please refer to the WOAH *Guide to Terrestrial Animal Health Surveillance*

	Type	Method	Definition*	When to consider	Advantages	Disadvantages	Suggested consideration for SHPS
Surveillance types	Passive surveillance (P)		Observer-initiated reporting of animal health data (e.g. voluntary disease notifications) or using existing data for surveillance. Data provision decisions are made by the data provider[1,4]	At any time	Low cost and high coverage	May lack sensitivity due to insufficient awareness and specificity due to non-pathognomonic clinical signs that are hard to differentiate from diseases with similar clinical presentations (e.g. Newcastle disease) unless confirmed by specific tests like immunodiffusion or real-time reverse transcription PCR. Requires new processes and financial support for communication and reporting	Training to recognise clinical signs can enhance specificity, especially when vaccine-preventable differential diagnoses are effectively controlled. A syndromic approach with passive surveillance that evaluates the overall health of poultry flocks and considers production indicators, such as sudden high poultry mortality or decreased egg production in hybrid layers, may be more effective and useful for SHPS, especially in resource-limited settings
Surveillance types	Active surveillance (A)		Investigator-led collection of animal health data using a set protocol for scheduled actions. The investigator decides what information to collect and from which animals[1,4]	At any time, particularly when the country is considered at risk	Generate more complete data in a timelier manner compared to passive surveillance	Resource-intensive, with increased personnel and financial requirements	Active surveillance is more practical in large-scale commercial poultry farms, especially those near SHPS. It can also be applied to high-risk groups, such as farmers, poultry traders and processing staff, as well as locations like farms and slaughterhouses, using a risk-based approach. Risk assessments can help identify high-risk areas within the smallholder poultry value chain
Surveillance methods	A	Participatory Disease Surveillance	The application of participatory appraisal methods (e.g. ranking, scoring or mapping) to disease surveillance. PDS primarily involves conducting semi-structured interviews with key informants, allowing communities to share knowledge about health events and risks, thereby gathering qualitative health data and establishing the presence or absence of a specific disease in a particular area. The goal is to improve case identification based on clinical definitions, which can later be validated with rapid field tests or laboratory diagnostics[1,2,4]	At any time, particularly when the country is considered at risk	Enables communities to share knowledge on health events and risks by gathering qualitative health data from specific populations	Participation and communication can be challenging in countries with limited capacity and resources. There may be a lack of sensitivity due to insufficient awareness and a lack of specificity because clinical signs can be mild and/or similar to other diseases (e.g. Newcastle disease) unless confirmed by specific tests	The effectiveness of the participatory surveillance system relies on the communication and trust established between the data farmer and the veterinarian or animal health worker
Surveillance methods	A	Sentinel surveillance	The ongoing collection of information from specific sites or groups (e.g. veterinary practices, laboratories, herds or animals) to track changes in the health status of a larger population over time. These sentinels serve as a proxy for the broader population and can be chosen based on risk, randomly or for convenience[1,4]	Affected or at-risk countries	The presence of clinical signs and mortality in sentinel animals can enhance HPAIV detection. Using unvaccinated sentinels to monitor vaccinated flocks may be more effective by focusing on a broader range of clinical signs rather than relying solely on flock-level detection thresholds	Detection of clinical signs is not a specific indicator of HPAIV infections and requires laboratory confirmation. False positives have occurred where sentinel chickens tested positive serologically, but virological tests failed to detect the virus. This can result from issues like mixing vaccinated and sentinel chickens, mis-vaccination, or cross-reactivity with other AIVs. Testing sentinel chickens also incurs additional costs	Unvaccinated chickens placed among vaccinated flocks can help detect AI virus outbreaks. The effectiveness of these sentinels is enhanced when they are located near the barn's air outlet. Additionally, sentinel surveillance in nearby wild bird populations, particularly domestic geese and ducks, may also aid in AI monitoring among migrating birds
Surveillance methods	A	Syndromic surveillance	Surveillance using health-related information (like clinical signs) can indicate potential health changes in a population, warranting further investigation or timely assessment of health threats. It is typically not hazard-specific, allowing it to detect various diseases, including emerging ones, and is especially useful for early warning purposes[1,4]	At any time, particularly when the country is considered at risk	Particularly applicable for early warning surveillance	Low specificity for HPAIV. Challenges to link suspected outbreaks to response actions, which require coordination of local personnel and may be resource-intensive. Lack of human and technological resources can affect data collection, management timeliness, and sharing	Instead of focusing on HPAIV, a syndromic surveillance approach that uses sudden/abnormal changes in morbidity, mortality or other production-related indicators (e.g. drop in egg production) may provide more practical and useful early warning systems for SHPS in resource-limited settings

*Definition of the surveillance types and methods was adapted based on Ameri *et al*. (2009), Hoinville *et al*. (2013), WOAH (2015) and FAO (2024):

[1] Hoinville L. Animal Health Surveillance Terminology Final Report from Pre-ICAHS Workshop. July 2013 (version 1.2), International Conference on Animal Health Surveillance. RISKSUR: 2013.
[2] Ameri AA, Hendrickx S, Jones B, Mariner J, Mehta P, Pissang C. Introduction to participatory epidemiology and its application to highly pathogenic avian influenza participatory disease surveillance: A manual for participatory disease surveillance practitioners. Nairobi (Kenya): International Livestock Research Institute; 2009.
[3] El Masry I, Delgado AH, Silva GOD, Lyons NA, Dhingra M. Recommendations for the surveillance of influenza A(H5N1) in cattle – With broader application to other farmed mammals. Rome (Italy): FAO; 2024.
[4] World Organisation for Animal Health (WOAH). Guide to Terrestrial Animal Health Surveillance. Paris (France): WOAH; 2015.

Table A1.1. (Cont.) Surveillance types and methods for consideration when designing a surveillance programme for High Pathogenicity Avian Influenza (HPAI) in smallholder poultry systems (SHPS) in resource-limited settings

The selection of surveillance types and methods should align with the established objectives of the programme and take into account the specific local context of each country. For additional guidance on surveillance types and methods, please refer to the WOAH *Guide to Terrestrial Animal Health Surveillance*

	Type	Method	Definition*	When to consider	Advantages	Disadvantages	Suggested consideration for SHPS
Surveillance methods	A	Routine inspection	Routine inspections consist of regular animal evaluations conducted by veterinary services throughout the year, irrespective of the HPAI epidemiological context. This process includes inspections at slaughterhouses, border checks, and quarantine measures for animals that fit the criteria for suspected cases[3,4]	At any time, particularly when the country is considered at risk	Takes advantage of routine activities to provide poultry samples for HPAIV detection. Can be a cost-effective way to target certain populations.	May require new processes and financial support for sample collection and shipment	Routine inspections of high-risk areas, such as slaughterhouses and live bird markets, could be a cost-effective way to conduct surveillance for small-scale poultry farms. Risk assessment can identify these high-risk nodes, enabling a more targeted, risk-based approach
	A	Opportunistic surveillance	A surveillance method that aims to take advantage of any existing animal health-related activities for the prevention and control of HPAIV in poultry to optimise the use of limited resources[3]	At any time, particularly when the country is considered at risk	Takes advantage of different animal-health-related or other health-related activities to observe or test poultry, may be low cost	May be less representative of the population of concern than other methods and require new processes and financial support for sample collection and shipment	Potential animal health-related activities that could allow for this surveillance approach may include, but are not limited to: • campaign for poultry vaccination for HPAIV or other poultry diseases (e.g. Newcastle disease, etc.) • procedures for surveillance and post-vaccination monitoring of other poultry diseases (e.g. Newcastle disease, etc.) The animal health teams responsible for these above mentioned activities should be vigilant to detect and report any suspected HPAIV cases matching the suspected case definition
	A	Disease survey	An investigation or study in which disease-specific information is collected systematically, usually carried out on a defined population group and within a defined time period[3,4]	At any time	Generate more complete data in the study population	Resource-intensive. Require new processes and financial support for sample collection and shipment	Adopt a risk-based surveillance approach to focus on animal populations having a higher risk of being infected with HPAIV, resulting in higher surveillance sensitivity with a lower sample size for higher efficiency and cost-effectiveness. • Surveys require a good understanding of the avian influenza virus epidemiology in the animal population (i.e. identification and quantification of risk factors) • Supply chain and value chain mapping, risk assessment, or risk-factor studies can be conducted
	P	Passive disease reporting	Observer-initiated provision of animal health related data (e.g. voluntary notification of suspect disease) or the use of existing data for surveillance. Decisions about whether information is provided, and what information is provided from which animals is made by the data provider[1,4]	At any time	Low cost and high coverage	May lack sensitivity due to low awareness and may be nonspecific because of vague clinical signs that overlap with other diseases (e.g. Newcastle disease) unless confirmed by immunodiffusion test or real-time reverse transcription PCR	Training to recognise clinical signs can enhance specificity. However, a syndromic approach that uses passive surveillance to assess the overall health of poultry flocks and considers production indicators (like decreased egg production in layers) may be more effective, especially in resource-limited settings
	Mixed (A,P)	Event-based (media-based, digital) surveillance	Surveillance that complements traditional indicator-based disease surveillance by continuously monitoring the internet and other media to identify emerging health threats. It relies on unstructured data that requires verification and cannot be summarised as indicators. This non-traditional approach serves as an early warning system, enhancing notifiable disease reporting and other health surveillance methods like sentinel or mortality surveillance[1,3]	At any time, particularly when the country is considered at risk	Can provide very high levels of coverage of a population at low cost or take advantage of other investments in event-based surveillance	Can be biased towards what is newsworthy rather than what is needed to meet the surveillance objectives. Detected events need to be confirmed using reliable sources of data	Event-based surveillance systems should be regularly monitored and evaluated to ensure effective early detection of health threats and identify areas for improvement

*Definition of the surveillance types and methods was adapted based on Ameri *et al.* (2009), Hoinville *et al.* (2013), WOAH (2015) and FAO (2024):

[1] Hoinville L. Animal Health Surveillance Terminology Final Report from Pre-ICAHS Workshop. July 2013 (version 1.2), International Conference on Animal Health Surveillance. RISKSUR: 2013.

[2] Ameri AA, Hendrickx S, Jones B, Mariner J, Mehta P, Pissang C. Introduction to participatory epidemiology and its application to highly pathogenic avian influenza participatory disease surveillance: A manual for participatory disease surveillance practitioners. Nairobi (Kenya): International Livestock Research Institute; 2009.

[3] El Masry I, Delgado AH, Silva GOD, Lyons NA, Dhingra M. Recommendations for the surveillance of influenza A(H5N1) in cattle – With broader application to other farmed mammals. Rome (Italy): FAO; 2024.

[4] World Organisation for Animal Health (WOAH). Guide to Terrestrial Animal Health Surveillance. Paris (France): WOAH; 2015.

Table A1.2. Suggested template for choosing surveillance types and methods (See **Table A1.1**) for High Pathogenicity Avian Influenza (HPAI) in smallholder poultry systems (SHPS) in resource-limited settings

The selection of surveillance types and methods should align with the established objectives of the programme and take into account the specific local context of each country. For additional guidance on surveillance types and methods, please refer to the WOAH *Guide to Terrestrial Animal Health Surveillance*.

Surveillance type (A = Active; P = Passive)	Surveillance method	Suggested list of surveillance attributes for consideration*								
		Overall feasibility and relevance	Population coverage	Surveillance sensitivity	Surveillance specificity	Time-liness	Cost (initial)	Cost (ongoing)	Inclu-siveness	Potential surveillance objective**
A	Participatory Disease Surveillance									
A	Sentinel surveillance									
A	Syndromic surveillance									
A	Routine inspection									
A	Opportunistic surveillance									
A	Disease survey									
P	Passive disease reporting									
Mixed (A,P)	Event-based (media-based, digital) surveillance									

* Qualitative assessment based on L (Low), M (Medium), or H (High)
** A – Early detection of HPAIV incursion in a specified SHPS population
B – Demonstration of freedom from HPAIV in a specified SHPS population
C – Describing the level of occurrence of HPAIV in a specified SHPS population
D – Detecting HPAIV cases in a specified SHPS population

Table A2. Suggested template for discussing the appropriate response measures for suspected, probable and confirmed High Pathogenicity Avian Influenza (HPAI) cases

Adapted from: Robyn M, et al. *Diagnostic Sensitivity and Specificity of a Participatory Disease Surveillance Method for Highly Pathogenic Avian Influenza in Household Chicken Flocks in Indonesia. Avian Dis. 2012;56(2):377-80.*

	To be provided by the Veterinary Services			To be provided by farmers			Final decision
	List of control measures	What are the advantages of this measure?	What are the disadvantages of this measure?	Are you considering adopting this measure?	What are the barriers to adopting this measure HERE in this local context?	Do you have any suggestions for enhancing adoption?	
Response measures for suspected cases	1...						
	2...						
	3...						
	4...						
	5...						
Response measures for probable cases	1...						
	2...						
	3...						
	4...						
	5...						
Response measures for confirmed cases	1...						
	2...						
	3...						
	4...						
	5...						

Table A3. Suggested template for the evaluation of the diagnostic procedures used in the surveillance programme

Adapted from: Robyn M, et al. *Diagnostic Sensitivity and Specificity of a Participatory Disease Surveillance Method for Highly Pathogenic Avian Influenza in Household Chicken Flocks in Indonesia. Avian Dis. 2012;56(2):377-80.*

Testing level	Results from each testing level							
Monitoring for clinical signs	Number of positive: …				Number of negative: …			
Rapid screening test	Number of positive: …		Number of negative: …		Number of positive: …		Number of negative: …	
Confirmatory test*	Number of positive: …	Number of negative: …	Number of positive: …	Number of negative: …	Number of positive: …	Number of negative: …	Number of positive: …	Number of negative: …

*For the full list of confirmatory laboratory tests recommended for HPAI, please refer to the WOAH *Manual of Diagnostic Tests and Vaccines for Terrestrial Animals*

Table A4. Suggested list of evaluation indicators for High Pathogenicity Avian Influenza (HPAI) surveillance in smallholder poultry systems (SHPS) in resource-limited settings

These indicators should be adapted and continuously updated to fit the unique contexts of different countries. This list of surveillance evaluation indicators should be used in conjunction with the WOAH *Guide to Terrestrial Animal Health Surveillance*, as well as the *RISKSUR EVA tool (Survtool)*. Adapted from Crawley *et al*. [36].

	Indicator	Indicator Description	Numerator	Denominator	Data Source
Impact	Reduced mortality	Mortality rate in a smallholder poultry population due to HPAIV under surveillance is reduced compared to the previous year(s)	Deaths in smallholder poultry population for HPAIV	Number of poultry in specified population	Surveillance reporting tools, all-cause mortality reports
	Reduced HPAI-specific mortality	Ratio betwen mortality in a smallholder poultry population due to HPAIV compared with other causes of mortality	Mortality in smallholder poultry population with mortality due to HPAIV	Mortality in smallholder poultry population due to causes other than HPAIV	Surveillance reporting tools
	Reduced spillover	Proportion of zoonotic events (initially detected in the animal population) that spilled over into, or led to human cases is reduced in comparison to previous year(s)	Number of zoonotic events detected in the animal population that spilled over into the human population	Total number of zoonotic events detected in the animal population	Surveillance reporting tools
	Reduced spread of events	Proportion of events that impacted >1 village (or the smallest administrative level) is reduced compared to previous year(s)	Number of events that spread to >1 village (or the smallest administrative level)	Number of all events reported	Surveillance reporting tools
	Reduced event response costs	Amount that event response costs are reduced compared to previous year(s)	Event response costs	N/A	Budgetary reports, costing analysis
	Cost-effectiveness of the surveillance programme	Cost effectiveness of detection and response using the surveillance programme	Cost of investments in the surveillance programme	Costs of detection and response activities for HPAIV under surveillance in jurisdictions not implementing the surveillance programme	Cost-effectiveness analysis
	Increased country support for the surveillance programme	Proportion of funding from the government for the surveillance programme has increased compared to prior year(s)	Funding allocated to the surveillance programme from government	Total national surveillance system budget	Budgetary reports, national surveillance system annual workplan
Outcomes	Usefulness and quality of surveillance data for decision-making	Proportion of decision-makers who responded with an average Likert scale rating on: (1) how **useful** the surveillance programme is for outbreak detection and response (2) whether the surveillance programme in their site/jurisdiction is **sensitive** enough (3) whether the surveillance programme in their site/jurisdiction is **specific** enough (4) how well the surveillance data are **trusted and considered accurate**	Number of decision-makers whose scores across four key criteria averaged ≥4	Total number of decision makers surveyed	Questionnaires, interviews, focus groups
	Sensitivity	**Case detection**: refers to the proportion of individual animals or herds in the population of interest that have the health-related condition of interest that the surveillance system is able to detect	Number of farms reported and confirmed with HPAI	Number of farms detected and confirmed with HPAI, plus those confirmed with HPAI but not detected by the surveillance programme	Surveillance reporting tools and databases
		Outbreak detection: refers to the probability that the surveillance system will detect a significant increase (outbreak) of disease. This may be an increase in the level of a disease that is not currently present in the population or the occurrence of any cases of disease that is not currently present	Number of detected and confirmed outbreaks	Number of detected and confirmed outbreaks, plus those not detected by the surveillance programme	
	Positive predictive value (PPV)	Probability that HPAIV is present (verified by confirmatory test) given that HPAIV is detected	Total number of confirmed HPAIV case	Total number of detected HPAIV cases	Surveillance reporting tools and databases

Table A4 (Cont.). Suggested list of evaluation indicators for High Pathogenicity Avian Influenza (HPAI) surveillance in smallholder poultry systems (SHPS) in resource-limited settings

These indicators should be adapted and continuously updated to fit the unique contexts of different countries. This list of surveillance evaluation indicators should be used in conjunction with the WOAH *Guide to Terrestrial Animal Health Surveillance*, as well as the *RISKSUR EVA tool (Survtool)*. Adapted from Crawley *et al*. [36].

	Indicator	Indicator Description	Numerator	Denominator	Data Source
Outcomes	Timeliness	**For early detection** – Measured using time – Time between introduction of infection and detection of outbreak – Measured using case numbers – Number of animals/farms infected when outbreak detected **For demonstrating freedom** – Measured using time – Time between introduction of infection and detection of presence by surveillance system – Measured using case numbers – Number of animals/farms infected when infection detected **For case detection to facilitate control** – Measured using time – Time between infection of animal/farm and their detection – Measured using case numbers – Number of other animals/farms infected before case detected **For detecting a change in prevalence** – Measured using time – Time between increase in prevalence and detection of increase – Measured using case numbers – Number of additional animals/farms infected when prevalence increase is identified			Surveillance reporting tools and databases
Outcomes	Surveillance programme utility	Proportion of community leaders and government stakeholders who responded with an average Likert scale rating on: (1) how **useful** the surveillance programme is for outbreak detection and response (2) whether the surveillance programme in their site/jurisdiction is **sensitive** enough (3) whether the surveillance programme in their site/jurisdiction is **specific** enough (4) how well the surveillance programme is **trusted**	Number of community leaders and government stakeholders whose scores across four key criteria averaged ≥4	Total number of community leaders and government stakeholders surveyed	Questionnaires; semi structured interviews including focus groups with all staff and supervisor/mentor interviews
Outputs	Surveillance workforce motivation	Proportion of surveillance workforce with improved motivation to implement surveillance functions	Number of surveillance workforce with improved motivation to implement surveillance functions	Total number of surveillance workforce surveyed	Questionnaires; semi structured interviews including focus groups with all staff and supervisor/mentor interviews
Outputs	Surveillance workforce competency	Proportion of surveillance staff with improved competency in analysis and interpretation of surveillance data for early warning surveillance	Number of surveillance staff with improved competency in analysis and interpretation of surveillance data	Total number of surveillance workforce assessed	Questionnaires; semi structured interviews including focus groups with all staff and supervisor/mentor interviews
Outputs	Surveillance workforce capacity	Proportion of trained surveillance staff actively engaged in surveillance activities	Number of trained surveillance staff actively engaged in surveillance activities	Total number of trained surveillance staff	Questionnaires; semi structured interviews including focus groups with all staff and supervisor/mentor interviews
Outputs	Surveillance workforce improved knowledge	Proportion of surveillance staff with improved knowledge and skills in surveillance from pre- to post-test	Number of surveillance staff with improved knowledge and skills from pre- to post-test	Total number of trained surveillance staffe	Pre- and post-test
Outputs	Personnel equipped to conduct surveillance	Proportion of personnel with the appropriate materials/resources (e.g. training on value chain mapping) at each level	Number of surveillance staff equipped	Total number of surveillance staff	Programme records; surveillance training reports

Table A4 (Cont.). Suggested list of evaluation indicators for High Pathogenicity Avian Influenza (HPAI) surveillance in smallholder poultry systems (SHPS) in resource-limited settings

These indicators should be adapted and continuously updated to fit the unique contexts of different countries. This list of surveillance evaluation indicators should be used in conjunction with the WOAH *Guide to Terrestrial Animal Health Surveillance*, as well as the *RISKSUR EVA tool (Survtool)*. Adapted from Crawley *et al*. [36].

	Indicator	Indicator Description	Numerator	Denominator	Data Source
Activities	Surveillance evaluation site visits are conducted	Proportion of sites implementing the surveillance programme where evaluation site visits are conducted (including data review, focus groups and key informant interviews, as appropriate)	Number of sites implementing surveillance evaluated	Total sites implementing surveillance	Surveillance annual workplan, surveillance evaluation report
	Surveillance signals are updated periodically	Frequency that surveillance signals are reviewed and/or updated	Number of times per year that surveillance signals are reviewed and/or updated at the national level	N/A	Programme records
	Surveillance reporting units are using digital systems	Proportion of reporting units using digital systems for the surveillance programme	Number of sites utilising digital systems for the surveillance programme	Total sites implementing the surveillance programme	Programme records
Inputs	Existence of surveillance signals for all sources/sites	Surveillance signals are defined for detection of HPAIV at all levels and in all settings	N/A	N/A	Surveillance evaluation report
	Surveillance implementation sites and sources identified	Administrative levels and surveillance types identified for surveillance implementation	N/A	N/A	Surveillance annual workplan, surveillance evaluation reports, surveillance reporting tools
	Surveillance technical guidelines and Standard Operating Procedures (SOP) are approved and available for use	Proportion of sites (by surveillance type and administration level) implementing surveillance that have the surveillance technical guidelines and SOPs available	Number of sites implementing surveillance with guidelines and SOPs	Total number of sites implementing surveillance	Guidelines, SOPs, surveillance evaluation reports
	National surveillance focal point established	National surveillance focal point established	N/A	N/A	Programme records
	Surveillance implementation workplan is available	National surveillance implementation workplan developed and available	N/A	N/A	Programme records
	Surveillance training materials are available	Surveillance training modules and training materials developed, approved and available for use	N/A	N/A	Surveillance training materials; programme records